BoundaryBalance: consistency and stability of weak Dirichlet finite elements

Abstract

Weak constraints introduce a numerical parameter whose role depends on the variational formulation. This reproducible study compares strong elimination, boundary penalty, and symmetric Nitsche enforcement for a dimensionless scalar membrane problem on triangular meshes. An interior Schur complement locates the mesh-specific positive-definiteness threshold. At eight subdivisions per side the measured threshold is $\gamma_* = 1.56489264$. A linear patch distinguishes consistency from finite-penalty bias. For a smooth manufactured field, Nitsche's last measured refinement orders are 1.9676 in L^2 and 0.9957 in the H^1 seminorm. Increasing the parameter reduces trace mismatch but can increase conditioning costs without reducing field error. These are numerical verification results for the stated model, not experimental validation of a physical membrane.

1 Question, model and assumptions

How much boundary enforcement is enough, and can a small boundary error hide an inaccurate interior field? We examine three treatments of the same essential boundary data, keeping mesh, quadrature and linear solver fixed. The work is an original implementation and a controlled comparison of established methods; it does not introduce a new boundary formulation.

For a membrane with constant isotropic tension scaled to unity, small transverse displacement satisfies

$$-\Delta u = f \quad \text{in } \Omega = (0,1)^2, \qquad u = g \quad \text{on } \Gamma = \partial\Omega. \tag{1}$$

The same scalar operator describes steady diffusion with unit conductivity. Geometry, load and displacement are dimensionless; no material calibration is asserted. Bending stiffness, finite deformation, inertia, contact and spatially varying tension are excluded. The imposed trace occupies the entire boundary.

Each square in an $n \times n$ Cartesian grid is split along its southwest to northeast diagonal. Continuous, piecewise affine functions form V_h; there are $(n+1)^2$ nodal unknowns and $2n^2$ triangles. On a boundary edge e we define $h_e = |e| = 1/n$. This convention differs from a triangle-diameter convention and must be retained when comparing penalty parameters.

2 Boundary formulations

Write $(\cdot,\cdot)_\Omega$ and $\langle\cdot,\cdot\rangle_e$ for volume and edge integrals. Strong elimination prescribes boundary nodal values and tests only with zero-trace functions. Penalty enforcement adds $\sum_e (\gamma/h_e)\langle u_h - g, v_h\rangle_e$ to the volume weak form. At finite γ it approximates a Robin boundary relation $\partial_n u + (\gamma/h_e)(u-g) = 0$, so it is generally inconsistent with the target Dirichlet solution even if that solution lies in V_h.

Symmetric Nitsche enforcement [1, 2] instead solves

$$a_\gamma(u_h, v_h) = (f, v_h)_\Omega - \langle \partial_n v_h, g\rangle_\Gamma + \sum_e \frac{\gamma}{h_e}\langle g, v_h\rangle_e, \tag{2}$$
$$a_\gamma(w, v) = (\nabla w, \nabla v)_\Omega - \langle v, \partial_n w\rangle_\Gamma - \langle \partial_n v, w\rangle_\Gamma + \sum_e \frac{\gamma}{h_e}\langle w, v\rangle_e.$$

Integration by parts shows consistency for smooth exact solutions. The two flux terms retain symmetry; the penalty term supplies stability when sufficiently large. Consistency alone does not guarantee a well-conditioned discrete solve.

2.1 A discrete stability threshold

With nodal basis ϕ_i, let $K_{ij} = (\nabla\phi_i, \nabla\phi_j)_\Omega$, $C_{ij} = \langle \phi_i, \partial_n\phi_j\rangle_\Gamma$, and $B_{ij} = \sum_e h_e^{-1}\langle \phi_i, \phi_j\rangle_e$. Set $D = K - C - C^T$ and partition nodes into interior i and boundary b. Because B acts only on boundary nodes and $D_{ii} = K_{ii}$ is positive definite, Schur complementation gives

$$S_0 = D_{bb} - D_{bi}K_{ii}^{-1}D_{ib}, \qquad \gamma_* = \lambda_{\max}(-S_0, B_{bb}). \tag{3}$$

Here B_{bb} is positive definite. In exact arithmetic $A_\gamma = D + \gamma B$ is positive definite if and only if $\gamma > \gamma_*$: its Schur complement is $S_0 + \gamma B_{bb}$. This criterion concerns the finite matrix, not a mesh-uniform continuum coercivity constant. Full-matrix eigenvalues provide a separate check of the condensation calculation on small meshes.

3 Implementation and reproducible experiments

The Python package assembles sparse matrices from analytic P1 gradients and outward normals of counterclockwise triangles. A six-point Gauss rule per Duffy coordinate integrates volume loads; six-point edge quadrature integrates boundary loads. The basis-product operators are integrated exactly to floating-point precision. Sparse LU solves the linear systems; dense symmetric generalized eigensolves act only on the boundary Schur complement. SciPy supplies these linear algebra routines [3]. The diagnostic is capped at 512 boundary nodes to bound its dense storage and computational cost. No timing claims are made.

The algorithm assembles K, C, B and the loads, eliminates strong constraints or forms the weak system, computes γ_* for Nitsche, rejects parameters within $10^{-8}\max(1, |\gamma_*|)$ of the threshold, solves, and evaluates residuals and continuum errors. Arrays are double precision. Tests check the divergence theorem, patch consistency, symmetry, invalid inputs, quadrature refinement and convergence.

The smooth benchmark uses

$$u = 1 + x + 2y + \sin(\pi x)\sin(\pi y), \qquad f = 2\pi^2\sin(\pi x)\sin(\pi y), \quad g = u|_\Gamma. \tag{4}$$

Refinement uses $n = 4, 8, 16, 32$ and $\gamma = 10$. The parameter sweep fixes $n = 8$ and samples 13 values between 1 and 10^5, including points on both sides of the threshold. The patch field is $0.7 + 1.3x - 2.1y$ with zero load. Error integrals use eight-point quadrature per coordinate, evaluated against the continuous exact field, not its nodal interpolant. Commands, JSON/CSV data, software versions and generated figures are included in the repository.

4 Measured validation results

For the linear patch at $n = 8$, $\gamma = 10$, the largest nodal errors are 8.882×10^{-16} for Nitsche and 2.682×10^{-2} for penalty enforcement. Thus the latter's error persists even when the exact field is representable. For the smooth problem, Table 1 reports continuum field and trace errors; Fig. 1 also includes strong elimination. At $n = 32$, Nitsche gives $\|u - u_h\|_{L^2} = 1.328 \times 10^{-3}$ and $|u - u_h|_{H^1} = 1.090 \times 10^{-1}$. Its last refinement orders approach the expected second and first orders for smooth P1 solutions. The penalty L^2 order over the same last two grids is 0.8599; this observed value is not an asymptotic theorem.

Table 1: Measured absolute errors at $\gamma = 10$. H^1 denotes the gradient seminorm; trace errors are integrated along the boundary.

Method	n	$L^2(\Omega)$	H^1 seminorm	$L^2(\Gamma)$
Penalty	4	5.298×10^{-2}	7.848×10^{-1}	1.374×10^{-1}
Nitsche	4	6.495×10^{-2}	8.021×10^{-1}	5.473×10^{-2}
Penalty	8	2.372×10^{-2}	4.248×10^{-1}	6.737×10^{-2}
Nitsche	8	1.957×10^{-2}	4.284×10^{-1}	1.327×10^{-2}
Penalty	16	1.316×10^{-2}	2.189×10^{-1}	3.369×10^{-2}
Nitsche	16	5.194×10^{-3}	2.173×10^{-1}	3.179×10^{-3}
Penalty	32	7.253×10^{-3}	1.110×10^{-1}	1.691×10^{-2}
Nitsche	32	1.328×10^{-3}	1.090×10^{-1}	7.757×10^{-4}

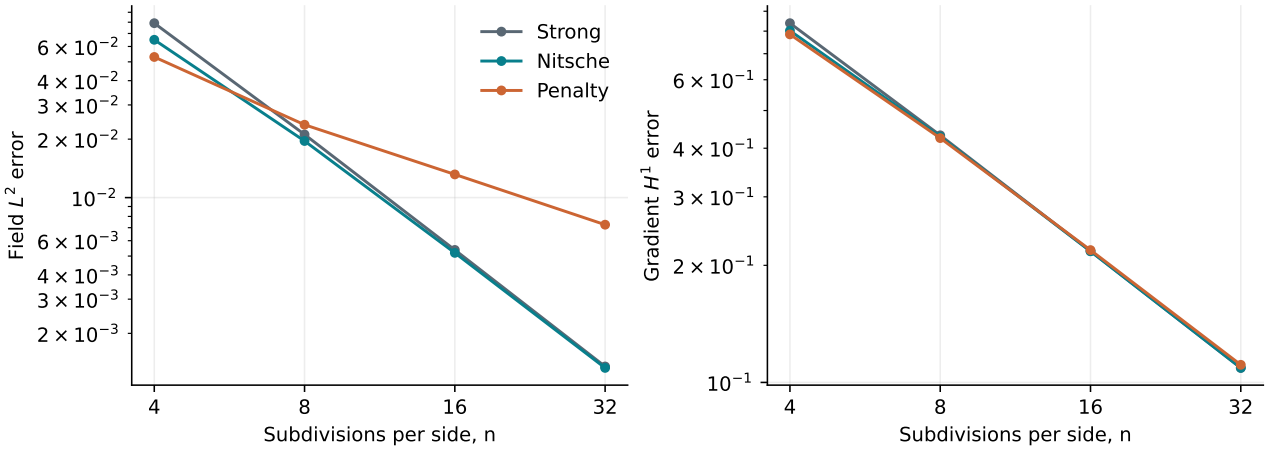


Figure 1: Refinement at fixed $\gamma = 10$. Strong and Nitsche results become close with refinement. Penalty consistency bias impairs the field-error trend even when gradient errors appear comparable. All curves are computed results.

4.1 Quadrature and algebraic error are separate

Increasing assembly quadrature from six to ten points per coordinate on $n = 4$ changes the maximum load entry by 3.025×10^{-13} and the computed Nitsche solution by 1.390×10^{-13}. Raising error quadrature from eight to twelve changes the largest reported norm by 7.772×10^{-16}. At $n = 8$, $\gamma = 10$, Nitsche's scaled backward indicator is 1.709×10^{-16} while its continuum L^2 error is 1.957×10^{-2}. The indicator is

$$\eta = \frac{\|Au_h - b\|_\infty}{\|A\|_\infty \|u_h\|_\infty + \|b\|_\infty}. \tag{5}$$

For strong enforcement it uses the reduced equations. A small η verifies algebraic satisfaction; it neither estimates continuum error nor certifies forward accuracy in a poorly conditioned system.

5 Parameter sensitivity and discussion

At $n = 8$ the condensed threshold is 1.56489264. Direct eigensolves at $0.999\gamma_*$ and $1.001\gamma_*$ confirm a sign change of the smallest full-matrix eigenvalue. The sweep excludes indefinite Nitsche matrices from error and condition-number plots. Even the stable choice $\gamma = 1.57$ gives L^2 error 3.314×10^{-1} and matrix condition number 2.082×10^3; crossing the threshold alone does not ensure useful accuracy.

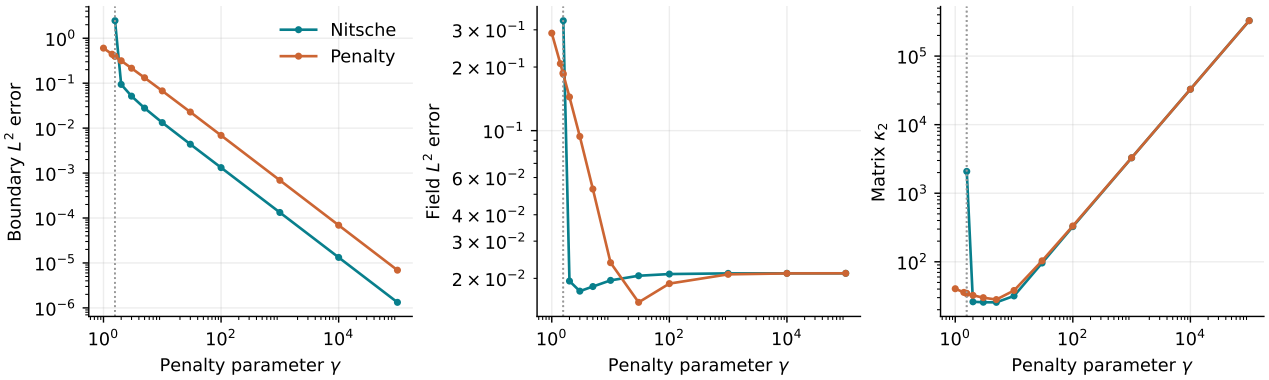


Figure 2: Parameter study on $n = 8$. Dotted lines mark γ_*. Only positive definite cases are plotted; κ_2 is the ratio of extreme eigenvalues of the unscaled full nodal matrix. The two methods approach strong enforcement as γ grows for this representable affine trace.

Larger penalties suppress trace error, but the field error reaches a discretization floor. At $\gamma = 10^5$, Nitsche's condition number is 3.284×10^5, without improved field accuracy. Penalty errors can also partially cancel spatial errors on a coarse grid: one small total error does not establish consistency or reliable convergence.

6 Limitations and conclusion

The Schur characterization holds in exact arithmetic; floating-point eigenvalues are not interval bounds. Boundary-dense matrices limit scalability. Condition numbers depend on basis scaling; no preconditioner is evaluated. Results concern one smooth problem, one affine patch, uniform diagonal meshes and unit coefficients. Singular solutions, distorted or cut elements, partial Dirichlet boundaries and heterogeneous media require further analysis.

Boundary mismatch should be assessed alongside consistency and continuum error. Nitsche recovers a linear patch and nearly second-order field convergence at a moderate parameter; finite penalty retains boundary bias. Neither accurate linear algebra nor manufactured-solution agreement establishes physical validity. The repository contains original code, measured data, figures and build checks, without course handouts, private datasets or experimental measurements.

References

[1] J. Nitsche. *Über ein Variationsprinzip zur Lösung von Dirichlet-Problemen bei Verwendung von Teilräumen, die keinen Randbedingungen unterworfen sind.* Abh. Math. Sem. Univ. Hamburg 36, 9–15 (1971). doi:10.1007/BF02995904.

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[3] P. Virtanen et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. Nature Methods 17, 261–272 (2020). doi:10.1038/s41592-019-0686-2.